Large-scale RNA-Seq Transcriptome Analysis of 4043 Cancers and 548 Normal Tissue Controls across 12 TCGA Cancer Types

Peng L., et al 2015

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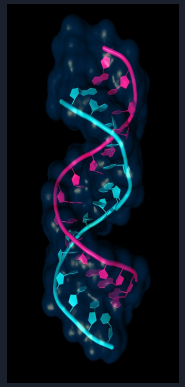
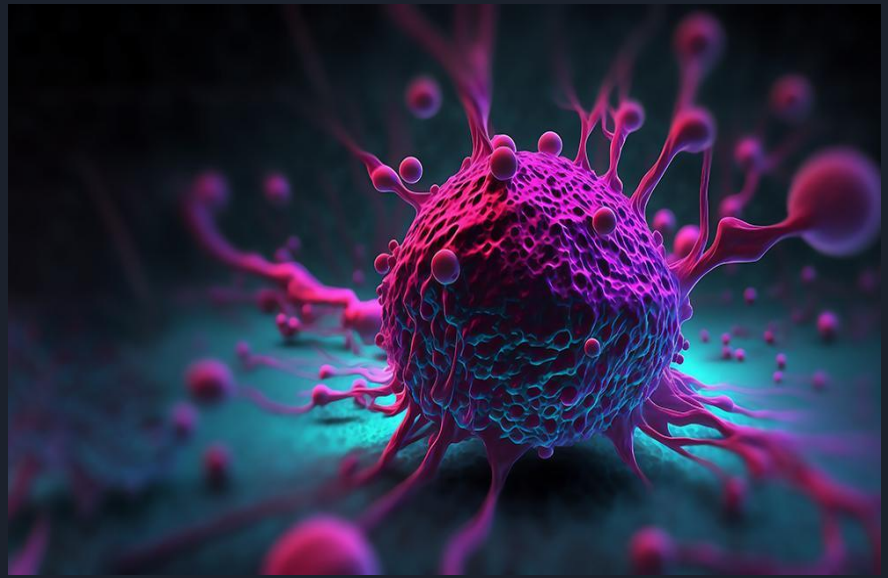
Introduction

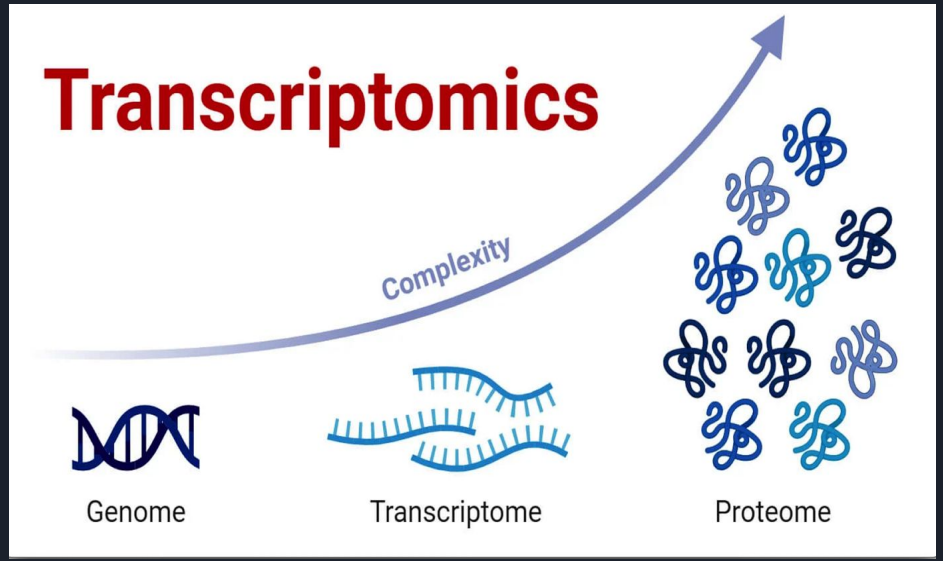


More than 100 types of cancer

RNA-seq data



Source: https://www.cancer.gov/types



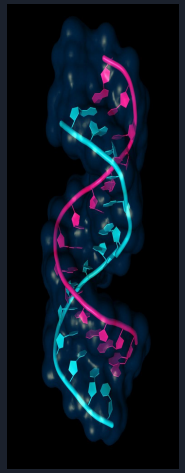
Source: https://microbenotes.com/transcriptomics/

Objectives

Analyze gene expression alterations on twelve different cancer types

RNA-seq data from the Cancer Atlas Genome (TCGA)

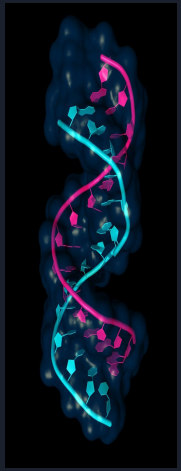
- Identify gene signatures that trigger tumorigenesis and metastasis



Methods

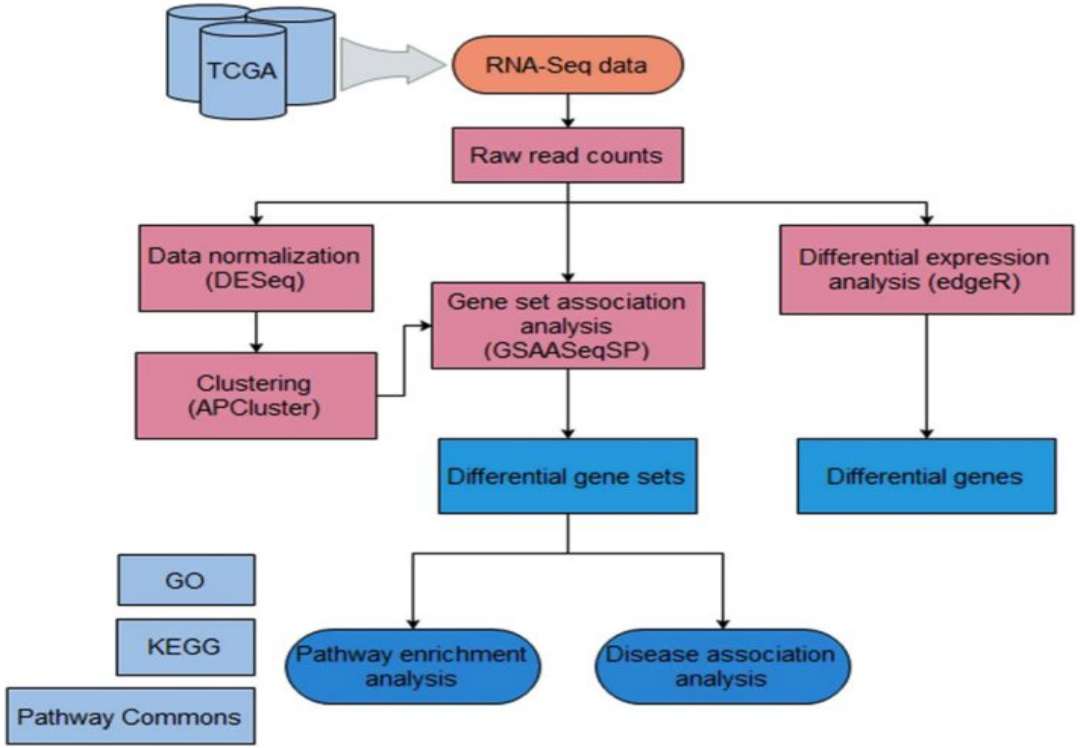
Analyzed **transcriptomes of 4043 cancer** and 548 normal tissue samples from **21 TCGA cancer types**

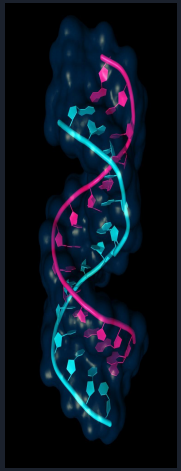
- Gene level DE analysis of transcriptome
- Gene clustering
- Gene set association analysis
- Cross cancer gene signatures
- Gene signatures



source:google.com

Workflow





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Key Findings

14-gene signatures that classified cancerous and normal samples across seven cancer types

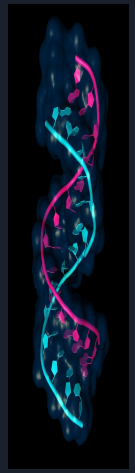
Seven cross-cancer gene signatures that were altered in at least four types of human cancers

Genes related to cell cycle regulation, common mechanism underlying tumorigenesis

Lung cancer-specific gene signatures

Gene signatures were validated by using non-TCGA datasets that can be used as potential biomarkers for cancer diagnosis and treatment.

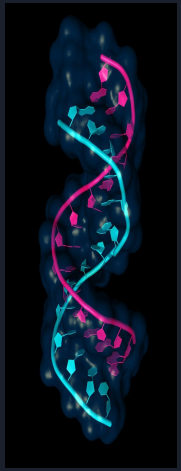
Our Final Interpretation



- **Cell cycle dysregulation** was highlighted as a common factor in many cancers, suggesting its key role in cancer development
- The **gene signatures were validated using independent datasets**, proving their effectiveness in distinguishing cancerous from normal tissues.
- These findings offer **potential for developing diagnostic tools and personalized therapies** tailored to the specific molecular features of different cancers.

source:google.com

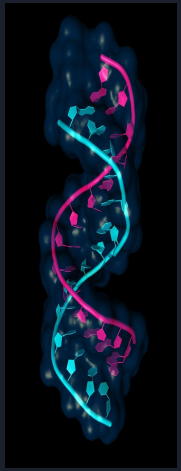
Challenges



Link between the **mutational profiles** and **differential expression (DE)** profiles in cancer

source:google.com

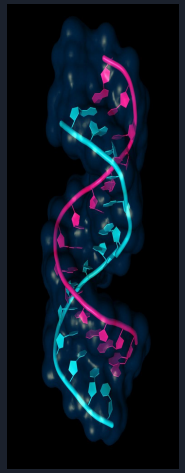
Future Research



- The relationship between mutational and DE profiles
- How gene expression patterns change in the vicinity of driver mutations

source:google.com

References



Peng, L., Bian, X. W., Li, D. K., Xu, C., Wang, G. M., Xia, Q. Y., & Xiong, Q. (2015). Large-scale RNA-Seq transcriptome analysis of 4043 cancers and 548 normal tissue controls across 12 TCGA cancer types. *Scientific reports*, *5*(1), 13413.

source:google.com

source:google.com